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परीक्षण प्रकोष्ठ स्थापित TESTING CELL ESTD. 1956

CSIR-CENTRAL INSTITUTE OF MINING AND FUEL RESEARCH

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FORMERLY : CENTRAL MINING RESEARCH INSTITUTE
(COUNCIL OF SCIENTIFIC & INDUSTRIAL RESEARCH)

बरवा रोड, धनवाद – 826015 (भारत) – BARWA ROAD, DHANBAD - 826015 (INDIA)

परीक्षण प्रमाण पत्र – *TEST CERTIFICATE*

FORMAT NO.: (CIMFR: DQM: FLP02: F-01:REV-01)
(Flame & Explosion Laboratory)

Prototype Report No.:	**CIMFR/TC/P/ 1526**	**Dated:**	**19th March,2014**
Equipment. ID NO.	**700/13**	**Code No.**	**FLP/219E/13-14**
Application Ref. No.:	**TFE/2012/2013-631**	**Dated:**	**06/12/2013**

1. **Applicant** : **M/S. TRI-FLP ENGINEERS PVT. LTD.,**
11, 1ST FLOOR, POPULAR INDL. ESTATE,
GOKHALE COMPOUND, RAGHUVIR NAGAR,
NEAR CHAR RASTA, DOMBIVALI EAST,
DIST – THANE 421 201 (MAHARASTRA)

2. **Manufacturer** : Same as above

3. **Apparatus** : FLAMEPROOF/WEATHERPROOF CONTROL PANEL.

4. **Designated by** : Cat. No.: TFE-CP-01

5. **Gas Group, Zone** : Zone 1& 2 and Gas Group : IIC atmosphere only.

6. **Electrical ratings:** rated at 250Amp. 220VAC/600VAC

7. **Temperature Class:** **(See Result#A2)**

8. **Degree of Ingress Protection:** IP-66 **(See Result#A4)**

9. **Material of Construction:** Cast Alumium Alloy LM-6 construction.

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10. Description of the apparatus: FLAMEPROOF/WEATHERPROOF CONTROL PANEL.

A.

Description	Volume in (ltrs.)		Min. Wall Thickness (mm)	Nos. & Size of Bolts/Fasteners
	Gross	Net		
First Enclosure	2.3	2	6mm	04 nos. M6x20mm long Allen cap screws with spring washer.
Second Enclosure	14.1	13		08 nos. M8x20mm long Allen cap screws with spring washer.
Third Enclosure	7.5	7		06 nos. M8x20mm long Allen cap screws with spring washer.

B.

Glass details		Max. no. of Aperture on cover
Second enclosure	**Third enclosure**	
Size = 175X175mm	Size = 105X105mm	**First Enclosure**
		1 no. glass window (8mm thick)
Shape = Square	Shape = Rectangular	Ø22.5mm for LED light
		1 no. push button &
Type = Toughened Type Glass	Type = Toughened Type Glass	1 no. rotary switch
		Second Enclosure
Thickness : 10mm,	Thickness : 10mm,	10 nos. glass window (8mm thick)
		Ø22.5mm for LED light
Cemented path : 15mm,	Cemented path : 15mm,	10 nos. push button
		3 nos. rotary switch.
Sealing Material : Epoxy Resin	Sealing Material : Epoxy Resin	**Third Enclosure**
		7 nos. glass window (8mm thick)
COT: - 20° C to 150°C	COT: - 20° C to 150°C	Ø22.5mm for LED light
		7 nos. push button
		1 no. rotary switch.

Note: glass is supported by a threaded backing ring.

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11. Nature of Flameproof Joint: Type of joints and gaps (Spigot joint, flange & threaded joint)

Sr. No.	Location of flamepath	Type of Joint	Min. length Axial threaded Length for threaded joint (in mm)		Max. gap (in mm) OR Min. No. of threads engaged and pitch size (in mm)	
			Req.	Spec.	Req.	Spec.
1.	Between first enclosure body and cover.	Spigot Joint	12.5	13.5	0.15	0.15
2.	Between second enclosure body and cover.	Spigot Joint	12.5	13.5	0.15	0.15
3.	Between third enclosure body and cover.	Spigot Joint	12.5	13.5	0.15	0.15

Push Button: Push button housing in brass is fitted into the enclosure cover maintaining 12.5(1.25P) axial threaded length. The push button spindle is fitted into the push button housing maintaining 13.5mm (min.) flamepath with max. dia. clearance is 0.15mm.

Rotary Switch: Brass bush is press fitted into the hole of cover maintaining 12.5mm (min.) flamepath. The spindle is fitted into the hole of bush maintaining 12.5mm (min.) flamepath and diametrical clearance is 0.15mm (max.).

Bushing: Brass bush 32mm φ is fitted into the hole of two enclosure. One end of bush is fitted by a threaded joint maintaining 15mm axial threaded length and other end is fitted by a cylindrical joint maintaining 15mm (min.) flamepath and diametrical clearance is 0.15mm (max.). Further it is locked by a check nut.

Max nos. of cable entries: Cable details entry .

Enclosure size in mm	M20	M25	M32	M40	M50
375X375X265	25	15	6	2	2
275X275X165	20	12	4	--	--
175X175X125	12	4	2	--	--

The above mentioned cable entries are provided on the enclosure body mentioned above maintaining 15mm axial threaded length with 1.5mm pitch for attachment of certified & approved double compression type cable gland.

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12. Name and Warning Inscription: The name-cum-warning inscription plate and rating plate made of SS/Brass is provided on the cover of enclosure. The warning inscription shall read as **"DO NOT OPEN WHEN ENERGIZED"**.

13. Drawings: The unit is designed and constructed as flameproof and weatherproof in accordance with the following drawing conforming to the requirements of IS/IEC 60079-0: 2004, IS/IEC 60079-1: 2007 and IS/IEC 60529:2001.

Sl. No.	Drg. Nos.	Title	Sheet	Date
1.	DRW-TFE-CP-01	Flameproof/Weatherproof Control Panel.	1 of 3	04/12/2013
2.	DRW-TFE-DB-01	Flameproof/Weatherproof Distribution Board.	2 of 3	06/12/2013
3.	DRW-TFE-SSC-03	Flameproof/Weatherproof Switch Socket Combine With Transformer.	3 of 3	06/12/2013

14. Any other relevant information:

i) As per clause no. 4.2.1 note-2 of IS/IEC60079-0:2004 the apparatus marked IIC is suitable for application requiring IIB apparatus.

ii) The material of construction of the control panel may be CI FG200 OR S.S/BRASS instead of Aluminum alloy LM-6.

iii) The above certified enclosure no. 2 may be used as Dol/star delta starter 75HP designated by cat. No.: TFE-SDS-01.

iv) The above certified enclosure no. 2 size 375x375x165 may be Coupling used as Distribution Board. 15 Way 400Amp., designated by Cat. No.: TFE-DB-01.

v) The above certified enclosure may be independently used as junction box and designated by cat. No.: TFE-JB-375, cat no. TFE-JB-275 and cat. No. TFE-JB-175 respectively.

vi) By changing the cover of the above certified enclosure no. 3 it may be used as FLP/WP switch socket plug with transformer (with test) designated by Cat. No.TFE-SSC-03

Flamepath for Plug & Socket:

Sr. No.	Location of flamepath	Type of Joint	Min. length Axial threaded Length for threaded joint (in mm)		Max. gap (in mm) OR Min. No. of threads engaged and pitch size (in mm)	
			Req.	Spec.	Req.	Spec.
1.	Between cover & socket cover.	Spigot Joint	12.5	13.5	0.15	0.15
2.	Between socket cover & socket holder.	Spigot Joint	12.5	14	0.15	0.15
3.	Between socket holder & socket holder guide.	Spigot Joint	12.5	18	0.15	0.15
4.	Between socket holder and plug holder.	Spigot Joint	12.5	15	0.15	0.15
5.	Between plug and plug holder.	Spigot Joint	6	9	0.2	0.15

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15. Declaration by the Applicant/Manufacturer:

(i) As to standards with which the apparatus complies in respect of:
   a) Electrical apparatus for Potentially Explosive atmosphere – General requirements: IS/IEC 60079-0: 2004
   b) As to flameproof protection IS/IEC 60079-1: 2007
   c) Ingress protection as per IS/IEC 60529:2001

(ii) Composition of the cast aluminum alloy LM-6 declared by the manufacturer: Composition of cast aluminum alloy LM-6 purported to be forming the material of construction of the enclosure has been declared by the manufacturer in the drawing having magnesium 0.1% (max).

However no sample of the alloy was drawn from the prototype enclosure for verifying its chemical composition declared by the manufacturer.

16. Documents /Sample Submitted :

i) Application form
ii) Drawings
iii) Prototype/test sample

17. Compliance of prototype or sample with documents: The test sample of electrical apparatus submitted for the type tests complies with the manufacturers documents referred above.

Note: CIMFR has however not checked and tested the compliance of the apparatus to any standard other than the above standards.

SCOPE OF THE TEST CERTIFICATE

The Test Certificate issued by CIMFR testifies that the apparatus has been found to comply with the definition of flameproof cum weatherproof APPARATUS contained in the relevant Standard specifications. They do not vouch for the quality of the equipment in any other respect.

This Institute reserves the right to review amend or withdraw this Test Report at any time if considered necessary in the interest of safety.

Naveen Kumar

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परीक्षण प्रकोष्ठ स्थापित-१९५६ TESTING CELL ESTD. 1956 CENTRAL INSTITUTE OF MINING & FUEL RESEARCH

FORMAT NO.: (CIMFR: DQM: FLP02: F-01: REV-01)

(Flame & Explosion Laboratory)

Prototype Report No. CIMFR/TC/P/ 1826 Dated: 19th March,2014

ID NO.: 700/13 CODE NO. FLP/219E/13-14

REPORT OF TEST

Date of Test: 19 & 20/12/2013

Test Equipment Used:

(i) Computer, Kistler Piezo Electric Transducer and Charge Amplifier for conducting Explosion Pressure Tests.
(ii) Fabricated set up for IP Testing.
(iii) Impact test setup
(iv) Thermal shock Test Machine.
(v) Temperature test set up
(vi) Hydraulic Testing Machine.

Result #A: Tests as per IS/IEC 60079-0: 2004:

Type Tests			
Clause	**Tests**	**Remarks**	**Results (Complies, P- Pass, NA-Not Applicable, F-Fail) References**
1	Scope		**Complies**
2	Normative references		**Complies**
3	Terms and definitions		**Complies**
4	Apparatus grouping and temperature classification	Group – IIC & T6	**Complies (Result#A2)**
5	Temperatures	-20°C to 40°C ambient	**Complies**
6	Requirements for all electrical apparatus		**Complies**
7	Non-metallic enclosures and non-metallic parts of enclosures	Tested Nylon as declared by manufacturer.	**Complies**
8	Enclosures containing light metals	For Gr. IIC Mg. 0.1% Zone 1 & 2	**Complies**
9	Fasteners		**Complies**
10	Interlocking devices		**Complies**
11	Bushings		**Complies**
12	Materials used for cementing/Sealing material	Epoxy resin COT: - 20° C to 150°C	**Complies**
13	Ex components	----------	**NA**
14	Connection facilities and terminal compartments		**Complies**
15	Connection facilities for earthing and bonding conductors		**Complies**
16	Entries into enclosures		**Complies**
17	Supplementary requirements for rotating electrical machines		**NA**

ID NO.: 700/13 CODE NO. FLP/219E/13-14

18	Supplementary requirements for switchgear		**NA**
19	Supplementary requirements for fuses		**NA**
20	Supplementary requirements for plugs and sockets		**Complies**
21	Supplementary requirements for luminaries		**NA**
22	Supplementary requirements for cap lights and hand lights		**NA**
23	Apparatus incorporating cells and batteries		**NA**
24	Documentation		**Complies**
25	Compliance of prototype or sample with documents		**Complies**
26	Type tests		**Complies**
26.3	**26.3: Tests in explosive mixtures**		
26.3	Tests for Flameproof (Ex 'd') protection	Verification for Compliance to Ex 'd' requirements as per IS/IEC 60079-1: 2007	**P/As per Result #B**
26.4	**Cl. 26.4 Test of enclosures**		
26.4.2	Test for resistance to Impact		**P/As per Result#A1**
26.4.3/2 6.4.4	Drop test		**NA** (not a handheld apparatus or portable device)
26.4.5	**Tests for the Degree of protection IP**		
26.4.5	IP of apparatus	Ability to prevent ingress of dust & water – IP-66 as per IS/IEC 60529:2001	**P/As per Result#A4**
26.5	**Thermal tests**		
26.5.1	Temperature measurement		**P/As per Result#A2**
26.5.2	Thermal shock test – for glass parts		**P/As per Result#A3**
26.6	**Torque Test for bushings**		
26.6	Torque test for bushings	As declaration by manufacturer	**Complies**
26.7	**Cl 26.7: Tests for non- metallic enclosures or of non-metallic parts of encl.**		
26.8	Thermal endurance to heat	Thermal requirements declared by manufacturer for Nylon COT: -20° C to 150°C	**Complies**
26.9	Thermal endurance to cold		
26.10	Resistance to light	----------	**NA**
26.11	Resistance to chemical agents for Gr.I	----------	**NA** (only applicable for Gr. I apparatus)
26.12	Earth continuity test via non-metallic enclosure		**NA**
26.13	Surface resistance test of parts of enclosures of Non metallic materials	----------	**NA**

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26.14	Charging test		**NA**
26.15	Measurement of Capacitance		**NA**
27	**Routine verification and tests**	**Manufacturer's responsibility:** Compliance to prototype of the product and physical measurement of gap and flamepath and hydraulic test at the pressure given in Result#B2 for 1min. for every sample.	
28	Manufacturer's responsibility	Conformity of the apparatus to the applicable standards	**Complies**
29	Marking		**Ex d Gr. IIC IP-66 T6**
30	Instructions	Instruction related to use, installation, maintenance, adjustment to be provided by the manufacturer along with the apparatus.	

Result#A1: 1kg mass of hardened steel fallen vertically on the surface of the Ex d aluminium Alloy LM-6 enclosure and toughened type 10mm thick glass to create impact energy for group II apparatus.

Result#A2: The surface temperature rise classification of the above apparatus has been determined at 110% of rated supply (250A, 220V AC) to the terminal mounted inside the enclosure and the surface temperature classification is considered as T6 at maximum ambient temperature of 40°C

Result#A3: The toughened type glass having 10mm thickness was subjected to thermal shock test at maximum operating temperature.

Result#A4: IP-Protection: Neoprene 'O' ring rubber gasket (3mm thick) are provided at the end of the each joint for weatherproof protection. Tests have been conducted as per IS/IEC 60529:2001 as tabulated below for IP-66 degree of protection

Sl. No	Type Tests			Results
1.	Dust Test (First Numeral 6	Clause 13.4& 13.6	No dust had accumulated inside the enclosure.	**Pass**
2.	Water Test (Second Numeral 6)	Clause 14.2.6	No water had accumulated inside the enclosure in.	**Pass**

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Result #B: Tests in explosive mixtures: Tests for type of protection Ex 'd' as per IS/IEC 60079-1:2007.

Type Tests			
Clause No.	**Description and relevant tests**	**Remarks**	**Results (Complies, P- Pass, NA-Not Applicable, F-Fail) References**
1	Scope		**Complies**
2	Normative references		**Complies**
3	Terms and definitions		**Complies**
4	Equipment grouping and temperature classification	Gr. IIC & T6	**Complies (Result#A2)**
5	Flameproof joints	Spigot Joint/threaded	**Complies**
6	Cemented joints	Epoxy resin COT: - 20° C to 150°C	**Complies**
7	Operating rods		**Complies**
8	Supplementary requirements for shaft and bearings		**NA**
9	Light transmitting parts other than glass	----------	**NA**
10	Breathing and draining devices which form part of a flameproof enclosure		**NA**
11	Fasteners, associated holes and blanking elements		**Complies**
12	Materials and mechanical strength of enclosures- materials inside the enclosures		**Complies**
13	Entries for flameproof enclosures		**Complies**
14	Verification and tests		**Complies**
15.1	**Tests of ability of the enclosure to withstand pressure**		
15.1.2	Determination of Explosion pressure (Reference Pressure)		**P/As per Result #B1**
15.1.3	Over Pressure Test		**P/As per Result #B2**
15.2	**Test for non-transmission of Internal ignition**		
15.2.1.1	Test for non-transmission of an Internal ignition		**P/As per Result #B4**
15.3	Reserved for future use		
15.4	Flame proof enclosures with breathing & draining devices		**NA**
16	Routine tests		**To be conducted by the manufacturer as per Result #B2**
17	Switchgear for Group I		**NA**
18	Lamp holders and lamp caps		**NA**
19.3	**Supplementary requirements for type tests**		
19.3.1.3	Flame erosion test		**P/As per Result #B3**
19.3.2	Flammability	As declared by manufacturer	**Complies**
20	Marking		**The warning inscription shall read as "DO NOT OPEN WHEN ENERGIZED"**

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Result # B1: Determination of Explosion pressure (Reference Pressure Test) *:

Gas mixture: Test with 8% Ethylene in air .

No. of tests: 3 nos. test with each enclosure.

Test Condition: Enclosure fitted with component and clause 15.1.2.1

Type of Test		Gas Group	Gas Mixture % in Air		No. of Tests
Preliminary Test (reference pressure)		IIC	8% Ethylene in air		Three
Test Ref. No.	Ignition	Gauge Position	Max. Pressure in Bar	Time of Pressure rise in millisecond	Remark
First Enclosure					
PPM/7001301	Body	Body	4.02	15.48	No evidence of distress
PPM/7001302	Body	Body	4.18	14.52	-do-
PPM/7001303	Body	Body	3.94	17.76	-do-
Second Enclosure					
PPM/7001304	Body	Body	3.76	16.79	No evidence of distress
PPM/7001305	Body	Body	3.81	15.00	-do-
PPM/7001306	Body	Body	3.99	15.12	-do-
Third Enclosure					
PPM/7001307	Body	Body	3.97	10.39	No evidence of distress
PPM/7001308	Body	Body	4.08	10.86	-do-
PPM/7001309	Body	Body	3.55	10.8	-do-
Third Enclosure (with switch socket & plug cover variation)					
PPM/7001310	Body	Body	4.07	14.76	No evidence of distress
PPM/7001311	Body	Body	3.95	15.18	-do-
PPM/7001312	Body	Body	4.15	14.34	-do-
Plug Enclosure					
PPM/7001313	Body	Body	3.51	7.14	No evidence of distress
PPM/7001314	Body	Body	3.98	7.02	-do-
PPM/7001315	Body	Body	4.06	7.02	-do-

**Results shown for the pressure time curve (enclosed) are the highest recorded value obtained in the test.*

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Result #B2: Overpressure Test (Static Method): Hydraulic pressure test is conducted as per Cl. 15.1.3.1 of IS/IEC 60079-1: 2007.

Enclosure	Test reference nos.	Over pressure (kg/cm^2) maintained for 1 min.	Remark
First Enclosure	OPM/7001316	10.0 kg/cm^2	No damage or deformation
Second Enclosure	OPM/7001317	10.0 kg/cm^2	No damage or deformation
Third Enclosure	OPM/7001318	10.0 kg/cm2	No damage or deformation
Third Enclosure (with switch socket & plug cover variation)	OPM/7001319	10.0 kg/cm2	No damage or deformation
Plug Enclosure	OPM/7001320	10.0 kg/cm2	No damage or deformation

Result #B3: Test for Erosion by flame:

Gas Mixture: Tests with 37% Hydrogen in air for Gas Gr. IIC

No. of Tests: 50 nos. tests in the Third Enclosure (Test Ref.: ERF/7001321 to ERF/7001370)

Test Condition: As per Clause 15.1.1.1 for Nylon.

Result #B4: Test for Non transmission of internal ignition:

Gas Mixture: Test with 37% Hydrogen in air for Gas Gr. IIC

No. of Tests: 5 nos. test with each enclosure.

Test Condition: Clause 15.2, Table 6 and 15.2.1.1 (No precompression required)

Name of Enclosure	Test Ref. Nos.	Gas Group	Gas Mixture % in Air	No. of Test	Result
First Enclosure	EIT/7001371 to EIT/7001375	IIC	37 % Hydrogen	Five	Pass
Second Enclosure	EIT/7001376 to EIT/7001380	IIC	37 % Hydrogen	Five	Pass
Third Enclosure	EIT/7001381 to EIT/7001385	IIC	37 % Hydrogen	Five	Pass
Third Enclosure (with switch socket & plug cover variation)	EIT/7001386 to EIT/7001390	IIC	37 % Hydrogen	Five	Pass
Plug Enclosure	EIT/7001391 to EIT/7001395	IIC	37 % Hydrogen	Five	Pass

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CONCLUSION: The **FLAMEPROOF/WEATHERPROOF CONTROL PANEL**, in cast aluminium alloy LM-6 construction designated by **Cat. No.: TFE-CP-01,** meets the test requirements for flameproofness as per IS/IEC 60079-1: 2007, general requirements as per IS/IEC 60079-0: 2004 and for weatherproof (IP-66 degree) as per IS/IEC 60529:2001. Hence, the above mentioned apparatus may be used in Zone 1 & 2 and Gas Group - IIC atmosphere only.

Reported By

Navin Kumar

(NAVIN KUMAR)
Technical Officer
Dated: 18th March,2014
Flame & Explosion Laboratory
Central Institute of Mining & Fuel Research, (CSIR)
(Erstwhile: Central Mining Research Institute)
Barwa Road, DHANBAD – 826 015,
(JHARKHAND) INDIA

Approved By

18/03/14

(A. K. SINGH)
Head of the Department